

4(a)	plasma cell ; bod B-lymphocyte / B-cell / B-plasma cells	1
4(b)	T → S → Q → R → P ;; allow one mark for correct order T, S, Q, R and only P out of sequence: • TPSQR • TSPQR • TSQPR	2
4(c)(i)	any two from: (so) unable to, synthesise / AW, nucleic acid / DNA / RNA I genetic material or cannot carry out, DNA replication / S phase / synthesis phase ; A semi-conservative replication <i>plus any one from:</i> unable to produce, sister / identical, chromatid(s) ; cannot pass checkpoint to progress to mitosis ; complementary base pairing cannot occur (in DNA replication) ; A purine-pyrimidine pairing cannot occur A idea of no complementary bases (for replication) allow ref. to complementary nucleotides	2

Question	Answer	Marks
4(c)(ii)	*if mp4 causes mp1, or comes before mp1 in chronological sequence, allow <u>only</u> mp4 any three from: 1 *changed (DNA), base / nucleotide, sequence ; 2 base substitution leading to changed amino acid or deletion / insertion, causing, frameshift / described ; R if <i>ref. to</i> is insertion / deletion / frameshift, of amino acids / to amino acid chain 3 premature / AW, STOP codon causing shortened, polypeptide (chain) / protein ; <i>context of changed codon from substitution or frameshift</i> 4 *changes / affects / disrupts / AW, primary structure / sequence of amino acids ; 5 change to <u>active site</u>, qualified ; I does not function e.g. no longer complementary to substrate altered, shape / (tertiary) structure, (compared to functional enzyme) as, tertiary structure of enzyme has changed <i>ref. to</i> catalytic site changed no enzyme-substrate complexes form	3
4(d)(i)	human immunodeficiency virus ; <i>must be spelled correctly</i> A human, immuno-deficiency / immuno deficiency, virus	1
4(d)(ii)	any two from: antibody molecules, are broken down / become degraded / AW ; A not long-lived / short-lived <i>allow context of destruction by phagocytes or removal from circulation</i> I 'flushed out' unless qualified passive, treatment / immunity / prevention ; A not active immunity <u>immune response</u> not triggered ; I immune system does not respond (B / T), memory cells not produced ; high risk individual may be, continually / regularly, exposed to HIV ; AW	2

Question	Answer	Marks
4(d)(iii)	<i>max 2 if response is clearly about a person who is already infected with HIV</i> <i>allow (envelope) glycoprotein for gp120</i> any three from: 1 monoclonal antibodies are specific ; A ref. to complementary shapes (of, antigen/epitope, and antibody) 2 more than one (HIV), antigen / glycoprotein, is involved in, attaching / binding / entering, host cells ; 3 better protection / more efficient / increases chances / AW, against the different strains / variants / types / AW ; A better able to cope with / AW, antibody, resistance / resistant strains but R ref. to antibiotic resistance I ref. to 'immune' 4 different strains (of HIV) may have (slightly) different shapes of, gp120 / envelope glycoprotein A idea that HIV mutates so, gp120 may vary / different antigens present / different surface receptors present or mixture of (monoclonal) antibodies to allow for each variant of gp120 ; bod different strains (of HIV) may have different antigens 5 AVP ; e.g. VRC01, (only) binds to / acts against, one site on gp 120 ora different monoclonal antibodies will bind at different sites on gp120 mixture of antibodies, (may) bind to / act against, more sites on gp 120 antibodies can attach to a number of sites on HIV to increase body (immune system) response	3